Project Report

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Github Repository: https://github.com/bernicelau430/Label-Organization-Categories-NLP

Problem, Background, and Context

There are three main levels that companies and organizations are classified within, industry, group, and business. With this project we set out to create a model that is able to correctly predict a company's industry, group, and/or business given only their name. This is an important task as many large datasets, like financial records or business directories, contain company names but lack structured information about what those companies actually do. Being able to automatically categorize companies could help with data organization, financial analysis, and improve search and recommendation systems. However, this problem is quite difficult, as in many cases the name of a company may have little to do with the industry that a company is in. For example, without any prior knowledge of Apple's products, it would be quite difficult to infer what industry they belong to.

Project Overview

Given a database of about 60,000 existing organizations with all categories determined, our program accepts the user-inputted name of an organization and uses NLP techniques to guess at the categories across the industry, group, and business levels corresponding to that organization. Our tech stack includes Python and other libraries like pandas and NLTK for the main program. Our app is hosted locally through React.

Motivations

As a data analyst and automation engineering intern at an investment advisory firm, Bernice recently worked on a project where she wrote Python scripts to statistically calculate and create data visualizations on stock performance by various classifications like sector, market, and region. In the dataset that Bernice worked with, the stocks were already classified through the company's extensive databases and other paid third-party resources, but the data isn't always going to be as clean for future projects in the industry. As such, Bernice is interested in learning how to utilize NLP to label categories, not just for stocks, but for companies and organizations so that she can improve on her data cleaning and data mapping skills.

Josh has recently worked on generating data for a classification model at his internship, so he thinks that it would be interesting to work on an actual classification model. He also thinks that the concept of NLP based classification is very intriguing and would like to further explore the intersection between NLP and more traditional ML techniques.

Data Sources, Tools, and Services

Data Description

A database of about 60,000 existing organizations with all categories (i.e. industry, group, business) determined was provided by Professor Foaad Khosmood. This data is in the form of 3 .tsv files. The features from these files are as follows:

Organizations.tsv (689560 rows x 12 cols)

Column Header	Description
oid	organization ID (positive int)
name	name of the organization (str)
oc_id	concept ID of the concept this organization falls under (negative int)
city	IGNORE
state	IGNORE
type	IGNORE
lastTouched	IGNORE
analysis_flag	IGNORE
lastTouched_ts	IGNORE
source	IGNORE
verified	IGNORE
org_type	IGNORE

OrgCategories.tsv (57652 rows x 5 cols)

Column Header	Description
oid	foreign key to Organizations (positive int)
category_level	either Business, Group, or Industry (str)
category	value for category_level (str)
source	IGNORE
lastTouched	IGNORE

OrgConcept.tsv (15334 rows x 12 cols)

Column Header	Description
oc_id	concept ID (negative int)
name	name of the concept (str)
acronym	acronym for name, if any (str)
canon_oid	oid of the org that the concept is based on (positive int)
state	IGNORE
validated	IGNORE
merged_into	IGNORE
to_be_deleted	IGNORE
lastTouched	IGNORE
UNSPECIFIED	IGNORE
UNSPECIFIED	IGNORE
UNSPECIFIED	IGNORE

Notably, the OrgCategories data is formatted with 3 entries for each organization, one for each potential value of category_level.

Data Cleaning

In order to clean the data, we first needed to clean up the tsv files themselves, as they had a few inconsistencies that made them more difficult to parse. We had to write a cleaner script that reconstructs the rows in the .tsv which were broken up by newline characters, allowing us to cleanly parse them into a dataframe.

Once the .tsv files were cleaned up, we parsed them into pandas dataframes. Then, we dropped all the unimportant columns from the dataframes. Next, in order to deal with the formatting of the OrgCategories dataframe, we pivoted it so that each oid had its own row with columns for the Business, Group, and Industry names as opposed to each category type having its own row. Once the OrgCategories dataframe was pivoted, we were able to finally join the tables together on their shared features (oid and oc_id) in order to create a single dataframe containing all of our data.

The dimensions of our final dataframe are 18,626 rows x 9 columns with the 9 attributes being 'oid', 'name_org', 'oc_id', 'Business', 'Group', 'Industry', 'name_concept', 'acronym', and 'canon_oid'.

No additional tools or services were used for the data sourcing, extraction, and cleaning process.

Methodology: KNN

Data Preparation

We used Python and Python with pandas to clean and structure the provided datasets, as described above. After constructing a unified dataset where each organization appears exactly once with a valid oc_id, corresponding concept, and complete labels for group, industry, and business categories, we removed organizations with missing or "Uncoded" category values and filtered out Industry categories with only one sample so we could implement stratified splitting while ensuring consistency and meaningful evaluation later on. The additional filtering yielded 5,449 organizations across 1,126 unique concepts for model training. From there, the filtered dataset was split using stratified sampling with 80% for training (4,352 samples and 20% for testing (1,097 samples), stratified by Industry to maintain class distribution.

Model Architecture

Rather than using a traditional supervised classifier such as logistic regression or SVM, we implemented a concept-based k-Nearest Neighbors (kNN) approach with similarity-weighted voting. We selected this method because organization names are inherently tied to concepts (oc_id), and each concept has predetermined Business, Group, and Industry labels. Predicting the concept first ensures that the three categories remain logically consistent, preventing contradictory combinations and enabling hierarchical consistency. Additionally, organization names are often short and sparse (e.g., "Apple Inc", "Goldman Sachs"). Similarity-based retrieval is particularly effective for such data, as organizations with similar names frequently belong to similar concepts. Moreover, kNN provides strong interpretability, since predictions can be directly explained by showing the most similar known organizations and their similarity scores. Our approach avoids training three separate multi-class classifiers and naturally handles class imbalance through similarity weighting.

Text Processing and Feature Engineering

Our advanced preprocessing pipeline begins with normalization, where all organization names are converted to lowercase and common legal entity terms are standardized (e.g. “Inc” becomes “incorporated”, “LLC” becomes “limited_liability_company”, “Corp” becomes “corporation”, etc). We then remove unnecessary special characters while preserving meaningful structural elements of the name. Next, we expand common business abbreviations such as “intl” to “international,” “tech” to “technology,” and “sys” to “systems” to reduce sparsity and improve consistency across similar names. Then, we apply a smart stopword removal process that eliminates generic language while intentionally preserving business-relevant terms such as “company”, “group”, “corporation”, “international”, “systems”, “services”, “technologies” and “solutions” since they might carry predictive value. After preprocessing, we tokenize the text using space-based tokenization.

For feature engineering, we incorporated domain-specific signals to enhance classification performance. First, we detected industry-related keywords such as technology, healthcare, financial, manufacturing, retail, energy, telecommunications, education, real estate, and transportation to capture strong sector indicators. We also identified organization type markers that signal corporate structure, such as “corporation,” “LLC,” or “limited.” Additionally, we detected geographic scope indicators such as “international” or “global.” These extracted features were then concatenated with the fully preprocessed organization name to create a richer textual representation before TF-IDF vectorization and similarity computation.

Initial Vectorization and Hyperparameters

For vectorization, we used TF-IDF to convert organization names into numerical feature vectors. We conducted systematic hyperparameter optimization to improve performance across multiple dimensions. Our initial configuration used max_features=5000, ngram_range=(1,3), min_df=2, max_df=0.8, k=7 neighbors, and a similarity threshold of 0.05.

Prediction Algorithm

During inference, a user enters an organization name through the React web interface, and the system applies the same preprocessing and feature engineering steps used during training. The processed text is then transformed into a numerical representation using the trained TF-IDF vectorizer. From here, the program computes cosine similarity between the resulting query vector and all training organization vectors. The top k=7 (subjectively chosen) most similar neighbors are retrieved, without applying any similarity threshold filtering. Then, we perform similarity-weighted voting, where each neighbor contributes its concept (oc_id) proportionally to its cosine similarity score. The concept with the highest total weighted similarity is selected as the predicted concept. The Business, Group, and Industry categories associated with that concept are returned as the final hierarchical prediction. For transparency and interpretability, the system also displays (up to) the top 3 most similar organizations to help users understand and assess the prediction.

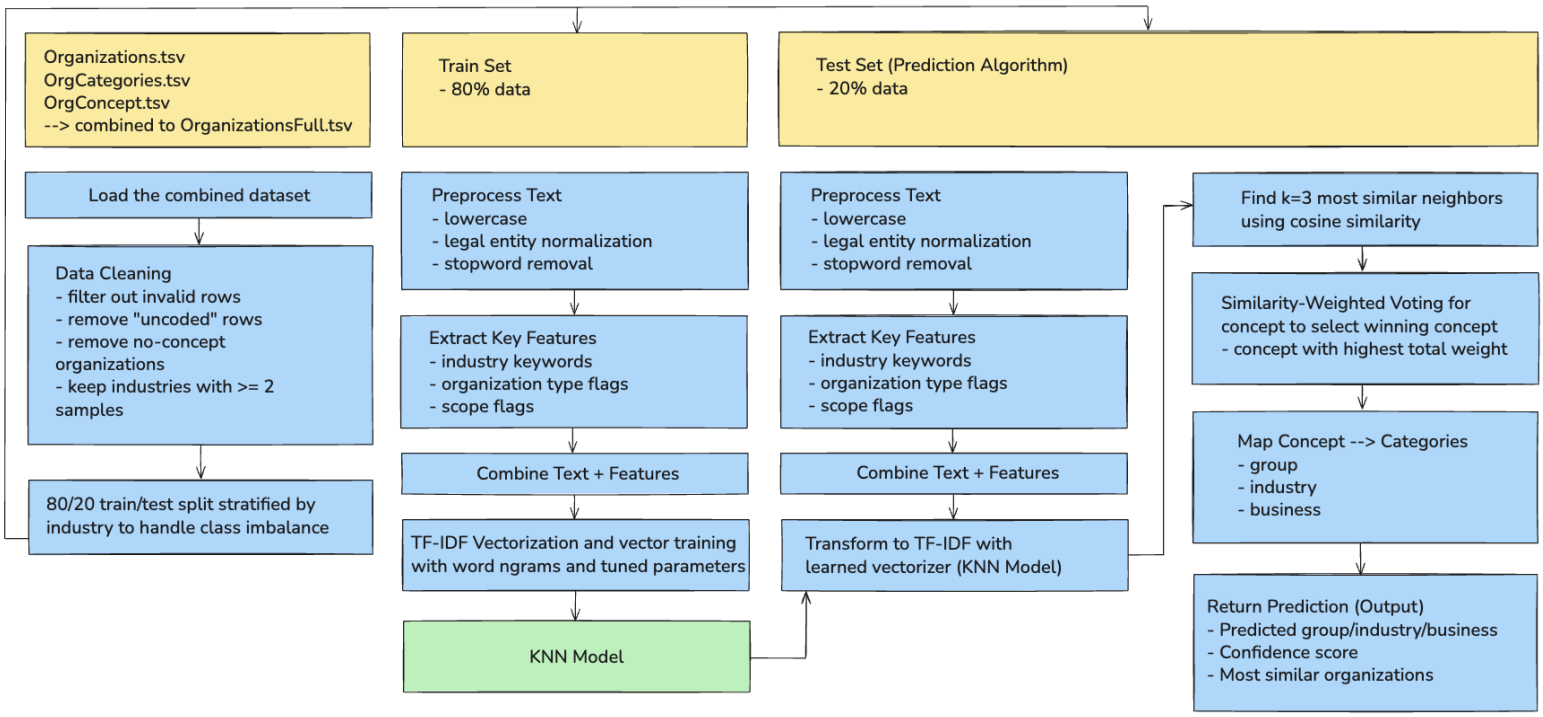

Fig 1. Design Diagram of the Base Model

Initial Evaluation

We measured performance using accuracy and macro-averaged F1 scores across the business, group, and industry category predictions, with an overall accuracy computed as the mean of the 3 category-level accuracies. The evaluation pipeline processed each test organization through the same preprocessing and feature extraction steps used during training, transformed the text into TF-IDF vectors using the learned vocabulary, retrieved the k-nearest neighbors from the training set via cosine similarity, and performed weighted voting to predict each label independently. We also tracked prediction confidence, which is derived from the similarity-weighted voting among the top neighbors. For each label, confidence was computed as the winning label's similarity weight divided by total neighbor weight. We computed confidence at the category level, as well as an overall confidence which was the mean of the confidence levels of each category. This allowed us to determine whether our predictions were certain or ambiguous, which is useful for deployment policies like flagging low-confidence cases for manual review. A key caveat is that these are relative voting confidences, not fully calibrated probabilities, so they should only be interpreted as ranking or uncertainty signals.

Base KNN Model Results

From this initial base model, we achieved an overall accuracy of 84.05%. The breakdown of accuracy and F1 scores by classification are detailed in Figure 2 and Appendix A, along with example runs shown in Appendix B.

	Business	Group	Industry
Accuracy	0.8090	0.8806	0.8320
Macro F1	0.6223	0.8568	0.6957

Fig 2. Accuracy and F1 Scores by Category for the Base KNN Model

Hyperparameter Tuning

To tune the hyperparameters of the model, we wrote a script to test every possible combination of the relevant hyperparameters. The relevant hyperparameters that we chose were the k value for the KNN algorithm, the word n-gram sizes for the tf-idf vectorizer, as well as the minimum and maximum values for dfs before we drop a term and whether or not the model uses word grams or character grams to make its tf-idf vectors. These configurations were set to optimize accuracy, and the optimal configuration was, k=1, word unigrams, with a min dfs of 1 and a max of 0.8. This model was further tweaked to a k value of 3 in order to avoid overfitting the dataset by using a larger value of k while only decreasing the accuracy score by <1%.

Tuned KNN Model Results

From the tuned model, we achieved an overall accuracy of 92.5%. The breakdown of accuracy and F1 scores by classification are detailed in Figure 3 and Appendix C, along with example runs shown in Appendix D.

	Business	Group	Industry
Accuracy	0.8990	0.9522	0.9229
Macro F1	0.7800	0.9321	0.8419

Fig 3. Accuracy and F1 Scores by Category for the Tuned KNN Model

Shortcomings, Failures and Problems of the Initial KNN Model

A notable concern from our model is that there could be data leakage since our training approach doesn't split by concept, meaning that while our dataset is comprehensive, our model may not generalize well on unseen concepts. The problem is that KNN is instance-based, so it doesn't learn patterns. Rather, it finds similar names in training data, so when the model encounters unseen concepts, it has no examples from those concepts to match against and ends up relying purely on name similarity to map to known concepts. Thus, our model can’t accurately classify organizations without a concept ID that it trained on. The model also doesn’t do much feature engineering in terms of handling extensive distinctions between common or abbreviated words like “inc” or “corp”. Moreover, the approach itself relies heavily on concepts to

map to outputs, which could lead to a trickle-down effect of bad predictions across all categories if the initial concept was incorrectly chosen.

Experimentation and Solutions

When we adapted our initial KNN model code to split by concept such that there was no concept overlap between the training and test sets, the overall accuracy dropped to 54.13%, as shown in Appendix E. We also attempted a Random Forest approach, which yielded 57.46% overall accuracy, as shown in Appendix F.

Upon further discussions with Professor Khosmood, we designed a data leakage-aware evolution of our earlier KNN approach, as outlined in Fig. 5, so that evaluation better reflects real deployment conditions rather than inflated accuracy from near-duplicate name overlap. It achieved an overall accuracy of 71.83%, as shown in Appendix G. The core implementation change was a group-level train/test split, where each row gets assigned a split_group_id using oc_id when available or a normalized-name fallback when oc_id is missing, and then the 80/20 split is performed on these groups with industry stratification at group level. This approach prevents lexical overlap between the train and test sets, while combating class imbalance. To support organizations without concept IDs, our revised approach no longer requires concept membership for training or inference and avoids concept-map-only prediction. Instead, it performs weighted KNN voting independently for Business, Group, and Industry using neighbor similarity scores, allowing no-concept organizations to be classified directly from learned textual patterns. To perform similarity-weighted voting for each label independently, the weighted_vote method loops through the k retrieved neighbors, accumulates their cosine similarity scores for each unique label value, and then selects the label with the highest accumulated score as the prediction. Like before, our UI displays the top 3 most similar organizations to help users understand and assess the prediction. Confidence is computed as the winning label's total score divided by the sum of all labels' scores, which represents how dominant that prediction was relative to alternatives. Feature engineering was also expanded to treat high-frequency legal and membership terms more deliberately. The model normalizes legal-entity variants like “inc”, “corp”, “llc”, etc, preserves selected business-critical tokens that might otherwise be removed as stopwords, and adds explicit symbolic indicators for organization-style cues such as “association”, “chapter”, and “union”, alongside existing industry and scope features. This combination combats data leakage-driven overestimation, broadens coverage to unlabeled/no-concept entities, and improves signal extraction from ambiguous organization names where generic terms appear across many classes. A built-in common-word behavior analysis further documents how pervasive terms distribute across industries, helping contextualize why some words are weak separators and why additional engineered features are necessary.

After this, we wanted to further improve our model’s performance when confidence values were low. Our previous model utilized word unigrams, which was very effective at classification when the input text contained word grams which our model was trained on. However, it failed to make any inference about company names that didn’t have any shared word grams. To fix this issue, we implemented a fallback model that was used when the confidence value produced by our initial model was below a certain threshold. This fallback

model was another KNN model that functioned off of character n-grams instead of word grams. This allowed our model to make inferences based on word structure instead of just entire words, allowing for a finer grain of classification. This had a minor impact on our test results, with the accuracy improving from 71.92% to 73.19%.

	Business	**Group**	**Industry**
Accuracy	0.6420	0.8185	0.7352
Macro F1	0.5471	0.8014	0.6691

Fig 4. Accuracy and F1 Scores by Category for the Revised KNN Model

Further Evaluation

In addition to evaluating accuracy, F1, and confidence scores as described earlier, we also conducted a common-word behavior analysis to quantify how generic terms like “union”, “group”, and “association” distribute across industries, which informed our feature engineering decisions and helped explain why certain words provide weak discriminative signal. This approach allowed us to compare our initial tuned baseline model against our revised model, while also assessing the system’s ability to classify organizations without predefined concept IDs.

Shortcomings, Failures and Problems of the Revised KNN Model

Remaining limitations are that KNN with TF-IDF still depends on lexical similarity and may struggle on truly novel naming patterns or sparse minority classes, but the revised design makes reported performance substantially more realistic and operationally relevant.

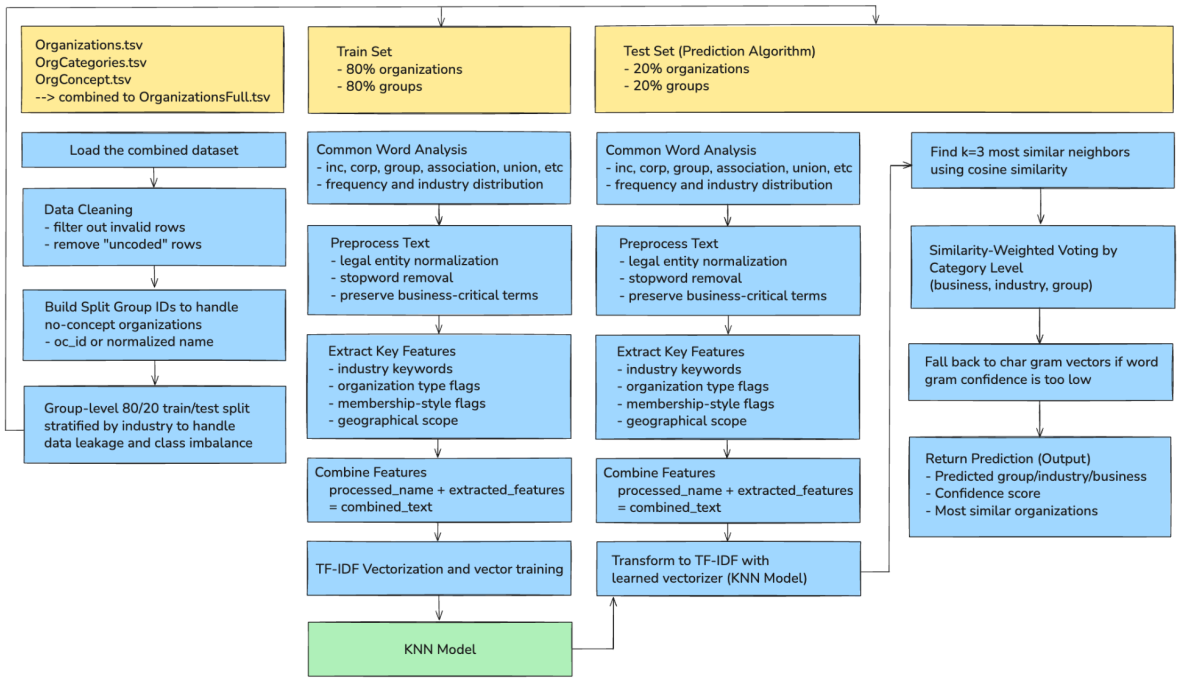

Fig 5. Design Diagram of the Revised KNN Model

Conclusion and Future Work

This project explored the problem of classifying organizations into their respective industry, group, and business using only their name. We developed a machine learning pipeline consisting of a TF-IDF vectorizer as well as a KNN model using both word and character n-grams. Our final, revised model showed strong performance on the testing set and demonstrated that it is possible to use NLP to extract useful data from the name of a company.

While the model performed well, there is still room for improvement to our pipeline, as in a real world environment 73% accuracy is not going to be good enough to be useful for most applications. To further improve our model, we could experiment with even more ML techniques such as support vector machines in order to try to improve our accuracy and F1. We could also improve our dataset to include more companies and expose our model to a greater variety of these names, allowing it to perform better Finally, our system could be extended beyond simple classification to include hierarchical prediction confidence or improved similarity explanations in order to make the model more useful overall.

References

- Pandas (Data Cleaning): https://dataanalysispython.readthedocs.io/en/latest/pandas.html
- Wordstream's Free Keyword Tool (Finding industry keywords): https://www.wordstream.com/keywords
- Preprocessing and NLP Techniques (Tokenization, Lemmatization, Normalization, Feature Engineering, NGrams): https://users.csc.calpoly.edu/~foaad/class/CSC482/slp3/

- Machine Learning Techniques (SVM, Logistic Regression, kNN): https://docs.google.com/presentation/d/1wljDk3pZxvvFCzUdYq9QZxSoKuLxH8awoV-lugiY5MA/edit?usp=sharing
- Machine Learning Techniques (Embeddings, TF-IDF, Cosine Similarity): https://docs.google.com/presentation/d/14bZlYHV3n6cvoNi9njkZMavHcgFXkJqWEcQ0ufBOhC8/edit?usp=sharing

Appendix

A. Baseline KNN Output

```
❯ python model_training.py
Loading data...
Total valid organizations: 5449
Unique concepts: 1126
Unique Business categories: 244
Unique Group categories: 15
Unique Industry categories: 94
After filtering rare categories: 5441 organizations
Industries removed: 8

Training set size: 4352
Test set size: 1089

==================================================
Training model with k=7...
==================================================
Preprocessing training data...
Training samples: 4352
Unique concepts: 1024
Creating TF-IDF vectors...
Model training complete!

Evaluating model...

Business Classification:
  Accuracy: 0.8090
  Macro F1: 0.6223

Group Classification:
  Accuracy: 0.8806
  Macro F1: 0.8568

Industry Classification:
  Accuracy: 0.8320
  Macro F1: 0.6957

Overall Accuracy: 0.8405
Mean Confidence: 0.7078
Model saved to ../model/org_categorizer_model.pkl
```

B. Example Output from Baseline KNN Model

```
==================================================
Test Example Predictions:
==================================================

Organization: Apple Inc
  Business: Employment agencies
  Group: General Business
  Industry: Business Services
  Similar organizations: ['Trueblue, Inc.', 'Draftkings Inc', 'Cadiz Inc.'
]

Organization: General Electric
  Business: Gas & electric utilities
  Group: Energy & Natural Resources
  Industry: Electric Utilities
  Similar organizations: ['Pacific Gas & Electric', 'Pacific Gas & Electri
c Corporation', 'Mercury General Corporation']

Organization: Microsoft Corporation
  Business: Computer software
  Group: Communications & Electronics
  Industry: Computer Equipment & Services
  Similar organizations: ['Microsoft Corporation', 'MICROSOFT CORP (FED PA
C)', 'Exxon Mobil Corporation']
```

C. Tuned KNN Model Output

```
❯ python model_training.py
Loading data...
Total valid organizations: 5449
Unique concepts: 1126
Unique Business categories: 244
Unique Group categories: 15
Unique Industry categories: 94
After filtering rare categories: 5441 organizations
Industries removed: 8

Training set size: 4352
Test set size: 1089

==================================================
Training model with k=3...
==================================================
Preprocessing training data...
Training samples: 4352
Unique concepts: 1024
Creating TF-IDF vectors...
Model training complete!

Evaluating model...

Business Classification:
  Accuracy: 0.8990
  Macro F1: 0.7800

Group Classification:
  Accuracy: 0.9522
  Macro F1: 0.9321

Industry Classification:
  Accuracy: 0.9229
  Macro F1: 0.8419

Overall Accuracy: 0.9247
Mean Confidence: 0.8721
Model saved to ../model/org_categorizer_model.pkl
```

D. Example Output from Tuned KNN Model

```
==================================================
Test Example Predictions:
==================================================

Organization: Apple Inc
  Business: Telecommunications
  Group: Communications & Electronics
  Industry: Telecom Services & Equipment
  Similar organizations: ['AT&T, Inc. and its Affiliates', 'At&T Inc And Its Affiliates', "AT&T Inc & It's Affiliates
"]

Organization: General Electric
  Business: Gas & electric utilities
  Group: Energy & Natural Resources
  Industry: Electric Utilities
  Similar organizations: ['Pacific Gas & Electric', 'Pacific Gas & Electric Corporation', 'GENERAL MOTORS COMPANY PAC
']

Organization: Microsoft Corporation
  Business: Computer software
  Group: Communications & Electronics
  Industry: Computer Equipment & Services
  Similar organizations: ['Microsoft Corporation', 'Microsoft Corporation PAC', 'MICROSOFT CORP PAC']
```

E. Tuned KNN Model (data split by concept)

```
❯ python model_training.py
Loading data...
Total valid organizations: 5449
Unique concepts: 1126
Unique Business categories: 244
Unique Group categories: 15
Unique Industry categories: 94
After filtering industries with <2 concepts: 5404 organizations
Valid concepts: 1111
Industries removed: 15

Training set: 4272 organizations, 888 concepts
Test set: 1130 organizations, 223 concepts
Verify no concept overlap: True

==================================================
Training model with k=3...
==================================================
Preprocessing training data...
Training samples: 4272
Unique concepts: 888
Creating TF-IDF vectors...
Model training complete!

Evaluating model...
Processed 2000/1130 samples...

Business Classification:
  Accuracy: 0.4310
  Macro F1: 0.2640

Group Classification:
  Accuracy: 0.6425
  Macro F1: 0.5147

Industry Classification:
  Accuracy: 0.5504
  Macro F1: 0.3504

Overall Accuracy: 0.5413
Mean Confidence: 0.6795
Model saved to ../model/org_categorizer_model.pkl
```

F. Random Forest Model (data split by concept)

```
❯ python random_forest_model.py
Loading data...
Total valid organizations: 5449
Unique concepts: 1126
Unique Business categories: 244
Unique Group categories: 15
Unique Industry categories: 94
After filtering industries with <2 concepts: 5404 organizations
Valid concepts: 1111
Industries removed: 15

Training set: 4272 organizations, 888 concepts
Test set: 1130 organizations, 223 concepts
Verify no concept overlap: True

================================================================
Training Random Forest Model
================================================================
Preprocessing training data for Random Forest model...
Training samples: 4272
Creating TF-IDF vectors...
Training Random Forest model for Business...
Training Random Forest model for Group...
Training Random Forest model for Industry...
Random Forest model training complete!

Evaluating Random Forest model...
Processed 2000/1130 samples...

Business Classification:
  Accuracy: 0.4717
  Macro F1: 0.2461

Group Classification:
  Accuracy: 0.6770
  Macro F1: 0.5484

Industry Classification:
  Accuracy: 0.5752
  Macro F1: 0.3316

Overall Accuracy: 0.5746
Model saved to ../model/org_categorizer_randomforest_model.pkl
```

G. Revised KNN Model

```
Loading data...
Total valid organizations: 8426
Orgs with concept id: 5449
Orgs without concept id: 2977
After filtering rare industries by group count: 8402 organizations
Industries removed: 12
Training set: 6589 organizations, 3159 groups
Test set: 1813 organizations, 790 groups
Leakage check - group overlap: 0

============================================================
Common Word Behavior Analysis
============================================================
incorporated    -> docs: 0 (0.00%), distinct_industries: 0, top_industry_share: 0.00
corporation     -> docs: 0 (0.00%), distinct_industries: 0, top_industry_share: 0.00
group           -> docs: 0 (0.00%), distinct_industries: 0, top_industry_share: 0.00
association     -> docs: 0 (0.00%), distinct_industries: 0, top_industry_share: 0.00
chapter         -> docs: 0 (0.00%), distinct_industries: 0, top_industry_share: 0.00
union           -> docs: 0 (0.00%), distinct_industries: 0, top_industry_share: 0.00
company         -> docs: 0 (0.00%), distinct_industries: 0, top_industry_share: 0.00
the             -> docs: 0 (0.00%), distinct_industries: 0, top_industry_share: 0.00

============================================================
Training model with k=3...
============================================================
Preprocessing training data...
Training samples: 6589
Unique split groups: 3159
Creating word TF-IDF vectors...
Creating char TF-IDF vectors...
Model training complete!

Evaluating model...
Processed 0/1813 samples...
Processed 1000/1813 samples...

Business Classification:
  Accuracy: 0.6420
  Macro F1: 0.5471

Group Classification:
  Accuracy: 0.8185
  Macro F1: 0.8014

Industry Classification:
  Accuracy: 0.7352
  Macro F1: 0.6691

Overall Accuracy: 0.7319
Mean Confidence: 0.8346
Model saved to ../model/org_categorizer_knn_model_v3.pkl
```